KIF5B fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes KIF5B gene fusions from the msk_impact_50k_2026 study, covering 96 fusion events. 85/87 fusions with known frame status (97.7%) were in-frame. 93/96 fusions (96.9%) retained the Kinesin motor domain. Domain retention was tested with Fisher's exact test (p=0.307781, not statistically significant at alpha=0.05). No literature benchmark is configured for KIF5B.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 96 analyzed fusion events, identifying 5 distinct fusion partner genes. The most frequent partner was RET, observed in 87 events.

Domain retention

Kinesin motor domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 83/85 (97.6%) of in-frame fusions retained the domain, p=0.307781 (not statistically significant at alpha=0.05). A breakpoint-position permutation test produced a corroborating empirical p-value of 9.9999e-06.

Domain disruption

Domain-disruption analysis was skipped: KIF5B has no disruption_required_domains configured; domain-disruption analysis was skipped.

Junction concentration

74% of 76 fusions with this gene as the 5' partner join at exon 15, against 12% expected if breakpoints fell at random along the gene; this is the most over-represented junction (empirical p=9.9999e-06, corrected for testing every exon, statistically significant at alpha=0.05). The random-breakage null assumes a breakpoint anywhere in the gene would be detected. Capture panels such as MSK-IMPACT tile only selected introns, so concentration on panel data is partly by assay design and this p-value overstates the evidence for biological concentration. Confirm in a cohort sequenced without intron selection (whole genome or RNA).

Cutpoint detection

Cutpoint detection scanned 79 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 575 aa (exon 15; label-permutation p=0.022977). This boundary is descriptive: retained/lost status follows from breakpoint position, so a clean split exists by construction and the p-value is not evidence of selection. This is 250 aa from the nearest configured domain boundary at 325 aa.

Corroborating confidence statistics

Corroborating confidence statistics compared Domain_retention_flags groups "retained" (n=76) and "not_retained" (n=3). For Frame_status, retained: 68/69 (98.6%, 95% CI 92.2%-99.7%); not_retained: 2/3

(66.7%, 95% CI 20.8%-93.9%). Welch's t-test comparing Tumor_variant_count between groups gave means of 53.7895 vs 43.3333, p=0.455031 (not statistically significant at alpha=0.05).

Composite evidence score

Composite evidence score reported: domain_disruption sub-score excluded: the supplied result had no applicable p-value (e.g. disruption_required_domains not configured for this gene).

Exon retention

Exon retention produced results for this run (1 row(s) in Exon_retention); see the accompanying table.

Expression association

Expression association reported: No mRNA expression data was available for KIF5B in this cohort; expression-association analysis was skipped.

Genomic position recurrence

Genomic position recurrence produced results for this run (12 row(s) in genomic_bin_recurrence, 1 row(s) in exact_position_recurrence, 4 row(s) in protein_position_genomic_spread); see the accompanying table.

Joint partner

Joint partner reported: KIF5B has no gene_pair configured; joint-partner analysis was skipped.

Mechanistic interpretation

Mechanistic interpretation produced results for this run; see the accompanying data for details.

Mutation cooccurrence

Mutation cooccurrence reported: KIF5B has no mutual_exclusivity_targets configured; co-occurrence/mutual-exclusivity analysis was skipped.

Window detection

Window detection produced results for this run (58 row(s) in window_scan, 23 row(s) in top_windows); see the accompanying table.

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
RET	87	0.90625	0.5000004342923104	None	0.8970121305869901	0.5970416227777926	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.7389138630486317	1
ARID5B	1	0.010416666666666666	0.5000004342923104	None	1.0	0.5970416227777926	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.42228658856357104	2
EGFR	1	0.010416666666666666	0.5000004342923104	None	1.0	0.5970416227777926	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.42228658856357104	3
ALK	5	0.052083333333333336	0.5000004342923104	None	0.0	0.5970416227777926	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.2504115885635711	4
MET	2	0.020833333333333332	0.5000004342923104	None	0.0	0.5970416227777926	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.23869283856357107	5

cutpoint_detection tables

cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
575	56	20	0	3	None	0.022395326192794548	1.6498426075796853
576	56	20	1	2	5.6	0.18597857838364168	0.7305370763495507
594	57	19	1	2	6.0	0.1708418164114367	0.7674058197477294
638	63	13	1	2	9.692307692307692	0.09073205275736923	1.04223926336373
639	63	13	2	1	2.423076923076923	0.44764096662830843	0.3490701751229587
698	64	12	2	1	2.6666666666666665	0.4213381555153707	0.3753692102786793
735	65	11	2	1	2.9545454545454546	0.3942133815551537	0.40426863770009347
848	68	8	2	1	4.25	0.3077808267681686	0.5117584380722447
849	68	8	3	0	0.0	1.0	-0.0

domain_retention tables

frame_domain_contingency_table

68	8
2	1

frame_domain_contingency_tables_by_tumor_type

stratum	table
AMPCA	[0, 1], [0, 0]
CSCLC	[1, 0], [0, 0]
CUP	[0, 1], [0, 0]
IMT	[1, 0], [0, 0]
LNET	[1, 0], [0, 0]
LUAD	[61, 6], [2, 1]
LUNE	[1, 0], [0, 0]
LUPC	[1, 0], [0, 0]
NSCLCPD	[2, 0], [0, 0]

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
auto_key_domain	Kinesin motor domain	79	76	0	3	0.0

exon_retention tables

Exon_retention

Exon	Retained_event_count	Total_event_count	Retained_fraction
1	76	79	0.9620253164556962

frequency tables

Partner_gene_counts

Partner_gene	Event_count	Raw_event_count	Sample_count	Patient_count
ALK	5	5	5	4
ARID5B	1	1	1	1
EGFR	1	1	1	1

Partner_gene	Event_count	Raw_event_count	Sample_count	Patient_count
MET	2	2	2	2
RET	87	87	86	71

genomic_position_recurrence tables

genomic_bin_recurrence

chromosome	bin_start	bin_end	n_events	event_ids
10	32315000	32316000	12	EVT-P-0003125-T01-IM5-26, EVT-P-0012648-T01-IM5-16, EVT-P-0014808-T01-IM6-18, EVT-P-0032227-T01-IM6-50, EVT-P-0032686-T01-IM6-52, ... (12 total)
10	32312000	32313000	11	EVT-P-0001010-T02-IM5-78, EVT-P-0002873-T01-IM3-13, EVT-P-0005573-T01-IM5-73, EVT-P-0008289-T01-IM5-11, EVT-P-0021038-T01-IM6-7, ... (11 total)
10	32314000	32315000	11	EVT-P-0002694-T02-IM5-12, EVT-P-0004995-T01-IM5-95, EVT-P-0011035-T01-IM5-15, EVT-P-0017523-T02-IM6-25, EVT-P-0019535-T02-IM6-28, ... (11 total)
10	32316000	32317000	11	EVT-P-0021578-T01-IM6-32, EVT-P-0021937-T01-IM6-34, EVT-P-0025928-T01-IM6-42, EVT-P-0027723-T01-IM6-47, EVT-P-0032753-T01-IM6-53, ... (11 total)
10	32313000	32314000	10	EVT-P-0000208-T01-IM3-51, EVT-P-0000252-T01-IM3-54, EVT-P-0009514-T01-IM5-14, EVT-P-0024843-T01-IM6-38, EVT-P-0026777-T01-IM6-45, ... (10 total)
10	32311000	32312000	8	EVT-P-0013825-T01-IM5-17, EVT-P-0018069-T01-IM6-27, EVT-P-0020273-T02-IM6-29, EVT-P-0032407-T01-IM6-48, EVT-P-0039474-T01-IM6-65, ... (8 total)
10	32304000	32305000	4	EVT-P-0005711-T02-IM6-2, EVT-P-0027991-T01-IM6-94, EVT-P-0043854-T01-IM6-9, EVT-P-0062804-T03-IM7-1
10	32305000	32306000	4	EVT-P-0008764-T01-IM5-3, EVT-P-0017924-T01-IM6-5, EVT-P-0024225-T01-IM6-37, EVT-P-0033090-T01-IM6-8
10	32306000	32307000	4	EVT-P-0017214-T01-IM6-23, EVT-P-0026693-T01-IM6-44, EVT-P-0043007-T01-IM6-71, EVT-P-0043704-T01-IM6-96

chromosome	bin_start	bin_end	n_events	event_ids
10	32317000	32318000	2	EVT-P-0016221-T01-IM6-21, EVT-P-0021150-T01-IM6-31
10	32309000	32310000	1	EVT-P-0064951-T01-IM7-92
10	32310000	32311000	1	EVT-P-0008101-T01-IM5-90

exact_position_recurrence

chromosome	position	n_events	event_ids
10	32315493	2	EVT-P-0012648-T01-IM5-16, EVT-P-0034357-T01-IM6-56

protein_position_genomic_spread

protein_position_aa	n_events	n_distinct_genomic_positions	genomic_span_bp	spans_multiple_chromosomes
575	56	55	4992	False
921	8	8	985	False
638	6	6	440	False
848	3	3	510	False

junction_concentration tables

junction_counts

role	junction_exon	count	role_event_count	observed_share	random_breakage_share	excess_neg_log10_p
five_prime	15	56	76	0.7368421052631579	0.11671797899708995	35.26031532879595
five_prime	24	8	76	0.10526315789473684	0.035848340432710554	2.230084304028769
five_prime	16	7	76	0.09210526315789473	0.01642697482181266	3.5816194607639678
five_prime	23	3	76	0.039473684210526314	0.01680654548521783	0.8655497582824533
five_prime	18	1	76	0.013157894736842105	0.0032896124161781452	0.6545707451699139
five_prime	19	1	76	0.013157894736842105	0.024714267639492218	0.07021696505950621
three_prime	16	1	3	0.3333333333333333	0.11766690565560288	0.5043261258604622
three_prime	17	1	3	0.3333333333333333	0.014929779427270043	1.3553254079121533
three_prime	24	1	3	0.3333333333333333	0.019168318501961115	1.2486460166135915

window_detection tables

window_scan

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
375	575	200	56	23	56	20	None	0.022395326192794548	1.6498426075796853	EVT-P-0000208-T01-IM3-51, EVT-P-0000252-T01-IM3-54, EVT-P-0001010-T02-IM5-78, EVT-P-0002694-T02-IM5-12, EVT-P-0002873-T01-IM3-13, ... (56 total)
376	576	200	57	22	56	20	5.6	0.18597857838364168	0.7305370763495507	EVT-P-0000208-T01-IM3-51, EVT-P-0000252-T01-IM3-54, EVT-P-0001010-T02-IM5-78, EVT-P-0002694-T02-IM5-12, EVT-P-0002873-T01-IM3-13, ... (57 total)
394	594	200	58	21	57	19	6.0	0.1708418164114367	0.7674058197477294	EVT-P-0000208-T01-IM3-51, EVT-P-0000252-T01-IM3-54, EVT-P-0001010-T02-IM5-78, EVT-P-0002694-T02-IM5-12, EVT-P-0002873-T01-IM3-13, ... (58 total)
438	638	200	64	15	63	13	9.692307692307692	0.09073205275736923	1.04223926336373	EVT-P-0000208-T01-IM3-51, EVT-P-0000252-T01-IM3-54, EVT-P-0001010-T02-IM5-78, EVT-P-0002694-T02-IM5-12, EVT-P-0002873-T01-IM3-13, ... (64 total)
439	639	200	65	14	63	13	2.423076923076923	0.44764096662830843	0.3490701751229587	EVT-P-0000208-T01-IM3-51, EVT-P-0000252-T01-IM3-54, EVT-P-0001010-T02-IM5-78, EVT-P-0002694-T02-IM5-12, EVT-P-0002873-T01-IM3-13, ... (65 total)
475	575	100	56	23	56	20	None	0.022395326192794548	1.6498426075796853	EVT-P-0000208-T01-IM3-51, EVT-P-0000252-T01-IM3-54, EVT-P-0001010-T02-IM5-78, EVT-P-0002694-T02-IM5-12, EVT-P-0002873-T01-IM3-13, ... (56 total)
476	576	100	57	22	56	20	5.6	0.18597857838364168	0.7305370763495507	EVT-P-0000208-T01-IM3-51, EVT-P-0000252-T01-IM3-54, EVT-P-0001010-T02-IM5-78, EVT-P-0002694-T02-IM5-12, EVT-P-0002873-T01-IM3-13, ... (57 total)
494	594	100	58	21	57	19	6.0	0.1708418164114367	0.7674058197477294	EVT-P-0000208-T01-IM3-51, EVT-P-0000252-T01-IM3-54, EVT-P-0001010-T02-IM5-78, EVT-P-0002694-T02-IM5-12, EVT-P-0002873-T01-IM3-13, ... (58 total)
498	698	200	66	13	64	12	2.6666666666666665	0.4213381555153707	0.3753692102786793	EVT-P-0000208-T01-IM3-51, EVT-P-0000252-T01-IM3-54, EVT-P-0001010-T02-IM5-78, EVT-P-0002694-T02-IM5-12, EVT-P-0002873-T01-IM3-13, ... (66 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
525	575	50	56	23	56	20	None	0.022395326192794548	1.6498426075796853	EVT-P-0000208-T01-IM3-51, EVT-P-0000252-T01-IM3-54, EVT-P-0001010-T02-IM5-78, EVT-P-0002694-T02-IM5-12, EVT-P-0002873-T01-IM3-13, ... (56 total)
526	576	50	57	22	56	20	5.6	0.18597857838364168	0.7305370763495507	EVT-P-0000208-T01-IM3-51, EVT-P-0000252-T01-IM3-54, EVT-P-0001010-T02-IM5-78, EVT-P-0002694-T02-IM5-12, EVT-P-0002873-T01-IM3-13, ... (57 total)
535	735	200	67	12	65	11	2.9545454545454546	0.3942133815551537	0.40426863770009347	EVT-P-0000208-T01-IM3-51, EVT-P-0000252-T01-IM3-54, EVT-P-0001010-T02-IM5-78, EVT-P-0002694-T02-IM5-12, EVT-P-0002873-T01-IM3-13, ... (67 total)
538	638	100	64	15	63	13	9.692307692307692	0.09073205275736923	1.04223926336373	EVT-P-0000208-T01-IM3-51, EVT-P-0000252-T01-IM3-54, EVT-P-0001010-T02-IM5-78, EVT-P-0002694-T02-IM5-12, EVT-P-0002873-T01-IM3-13, ... (64 total)
539	639	100	65	14	63	13	2.423076923076923	0.44764096662830843	0.3490701751229587	EVT-P-0000208-T01-IM3-51, EVT-P-0000252-T01-IM3-54, EVT-P-0001010-T02-IM5-78, EVT-P-0002694-T02-IM5-12, EVT-P-0002873-T01-IM3-13, ... (65 total)
544	594	50	58	21	57	19	6.0	0.1708418164114367	0.7674058197477294	EVT-P-0000208-T01-IM3-51, EVT-P-0000252-T01-IM3-54, EVT-P-0001010-T02-IM5-78, EVT-P-0002694-T02-IM5-12, EVT-P-0002873-T01-IM3-13, ... (58 total)
550	575	25	56	23	56	20	None	0.022395326192794548	1.6498426075796853	EVT-P-0000208-T01-IM3-51, EVT-P-0000252-T01-IM3-54, EVT-P-0001010-T02-IM5-78, EVT-P-0002694-T02-IM5-12, EVT-P-0002873-T01-IM3-13, ... (56 total)
551	576	25	57	22	56	20	5.6	0.18597857838364168	0.7305370763495507	EVT-P-0000208-T01-IM3-51, EVT-P-0000252-T01-IM3-54, EVT-P-0001010-T02-IM5-78, EVT-P-0002694-T02-IM5-12, EVT-P-0002873-T01-IM3-13, ... (57 total)
569	594	25	58	21	57	19	6.0	0.1708418164114367	0.7674058197477294	EVT-P-0000208-T01-IM3-51, EVT-P-0000252-T01-IM3-54, EVT-P-0001010-T02-IM5-78, EVT-P-0002694-T02-IM5-12, EVT-P-0002873-T01-IM3-13, ... (58 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
575	600	25	58	21	57	19	6.0	0.1708418164114367	0.7674058197477294	EVT-P-0000208-T01-IM3-51, EVT-P-0000252-T01-IM3-54, EVT-P-0001010-T02-IM5-78, EVT-P-0002694-T02-IM5-12, EVT-P-0002873-T01-IM3-13, ... (58 total)
575	625	50	58	21	57	19	6.0	0.1708418164114367	0.7674058197477294	EVT-P-0000208-T01-IM3-51, EVT-P-0000252-T01-IM3-54, EVT-P-0001010-T02-IM5-78, EVT-P-0002694-T02-IM5-12, EVT-P-0002873-T01-IM3-13, ... (58 total)
575	675	100	65	14	63	13	2.423076923076923	0.44764096662830843	0.3490701751229587	EVT-P-0000208-T01-IM3-51, EVT-P-0000252-T01-IM3-54, EVT-P-0001010-T02-IM5-78, EVT-P-0002694-T02-IM5-12, EVT-P-0002873-T01-IM3-13, ... (65 total)
575	775	200	67	12	65	11	2.9545454545454546	0.3942133815551537	0.40426863770009347	EVT-P-0000208-T01-IM3-51, EVT-P-0000252-T01-IM3-54, EVT-P-0001010-T02-IM5-78, EVT-P-0002694-T02-IM5-12, EVT-P-0002873-T01-IM3-13, ... (67 total)
576	676	100	9	70	7	69	0.050724637681159424	0.03292909622023547	1.4824201888736057	EVT-P-0003125-T01-IM5-26, EVT-P-0013825-T01-IM5-17, EVT-P-0018069-T01-IM6-27, EVT-P-0020273-T02-IM6-29, EVT-P-0032407-T01-IM6-48, ... (9 total)
576	776	200	11	68	9	67	0.06716417910447761	0.04938099874808736	1.3064401305564408	EVT-P-0003125-T01-IM5-26, EVT-P-0008101-T01-IM5-90, EVT-P-0013825-T01-IM5-17, EVT-P-0018069-T01-IM6-27, EVT-P-0020273-T02-IM6-29, ... (11 total)
588	638	50	7	72	7	69	None	1.0	-0.0	EVT-P-0013825-T01-IM5-17, EVT-P-0020273-T02-IM6-29, EVT-P-0032407-T01-IM6-48, EVT-P-0039474-T01-IM6-65, EVT-P-0048968-T01-IM6-77, ... (7 total)
589	639	50	8	71	7	69	0.2028985507246377	0.277241745596176	0.5571413753695974	EVT-P-0013825-T01-IM5-17, EVT-P-0018069-T01-IM6-27, EVT-P-0020273-T02-IM6-29, EVT-P-0032407-T01-IM6-48, EVT-P-0039474-T01-IM6-65, ... (8 total)
594	644	50	8	71	7	69	0.2028985507246377	0.277241745596176	0.5571413753695974	EVT-P-0013825-T01-IM5-17, EVT-P-0018069-T01-IM6-27, EVT-P-0020273-T02-IM6-29, EVT-P-0032407-T01-IM6-48, EVT-P-0039474-T01-IM6-65, ... (8 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
594	694	100	8	71	7	69	0.2028985507246377	0.277241745596176	0.5571413753695974	EVT-P-0013825-T01-IM5-17, EVT-P-0018069-T01-IM6-27, EVT-P-0020273-T02-IM6-29, EVT-P-0032407-T01-IM6-48, EVT-P-0039474-T01-IM6-65, ... (8 total)
594	794	200	10	69	9	67	0.26865671641791045	0.3374473627638185	0.47179396163015375	EVT-P-0008101-T01-IM5-90, EVT-P-0013825-T01-IM5-17, EVT-P-0018069-T01-IM6-27, EVT-P-0020273-T02-IM6-29, EVT-P-0032407-T01-IM6-48, ... (10 total)
598	698	100	8	71	7	69	0.2028985507246377	0.277241745596176	0.5571413753695974	EVT-P-0008101-T01-IM5-90, EVT-P-0018069-T01-IM6-27, EVT-P-0020273-T02-IM6-29, EVT-P-0032407-T01-IM6-48, EVT-P-0039474-T01-IM6-65, ... (8 total)
613	638	25	6	73	6	70	None	1.0	-0.0	EVT-P-0020273-T02-IM6-29, EVT-P-0032407-T01-IM6-48, EVT-P-0039474-T01-IM6-65, EVT-P-0048968-T01-IM6-77, EVT-P-0052861-T02-IM6-80, ... (6 total)
614	639	25	7	72	6	70	0.17142857142857143	0.2458174736655749	0.6093872490052457	EVT-P-0018069-T01-IM6-27, EVT-P-0020273-T02-IM6-29, EVT-P-0032407-T01-IM6-48, EVT-P-0039474-T01-IM6-65, EVT-P-0048968-T01-IM6-77, ... (7 total)
635	735	100	9	70	8	68	0.23529411764705882	0.3077808267681686	0.5117584380722447	EVT-P-0008101-T01-IM5-90, EVT-P-0018069-T01-IM6-27, EVT-P-0020273-T02-IM6-29, EVT-P-0032407-T01-IM6-48, EVT-P-0039474-T01-IM6-65, ... (9 total)
638	663	25	7	72	6	70	0.17142857142857143	0.2458174736655749	0.6093872490052457	EVT-P-0018069-T01-IM6-27, EVT-P-0020273-T02-IM6-29, EVT-P-0032407-T01-IM6-48, EVT-P-0039474-T01-IM6-65, EVT-P-0048968-T01-IM6-77, ... (7 total)
638	688	50	7	72	6	70	0.17142857142857143	0.2458174736655749	0.6093872490052457	EVT-P-0018069-T01-IM6-27, EVT-P-0020273-T02-IM6-29, EVT-P-0032407-T01-IM6-48, EVT-P-0039474-T01-IM6-65, EVT-P-0048968-T01-IM6-77, ... (7 total)
638	738	100	9	70	8	68	0.23529411764705882	0.3077808267681686	0.5117584380722447	EVT-P-0008101-T01-IM5-90, EVT-P-0018069-T01-IM6-27, EVT-P-0020273-T02-IM6-29, EVT-P-0032407-T01-IM6-48, EVT-P-0039474-T01-IM6-65, ... (9 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
638	838	200	9	70	8	68	0.23529411764705882	0.3077808267681686	0.5117584380722447	EVT-P-0008101-T01-IM5-90, EVT-P-0018069-T01-IM6-27, EVT-P-0020273-T02-IM6-29, EVT-P-0032407-T01-IM6-48, EVT-P-0039474-T01-IM6-65, ... (9 total)
648	848	200	5	74	5	71	None	1.0	-0.0	EVT-P-0008101-T01-IM5-90, EVT-P-0017214-T01-IM6-23, EVT-P-0026693-T01-IM6-44, EVT-P-0043007-T01-IM6-71, EVT-P-0064951-T01-IM7-92
649	849	200	6	73	5	71	0.14084507042253522	0.21349536539409958	0.6706115482999183	EVT-P-0008101-T01-IM5-90, EVT-P-0017214-T01-IM6-23, EVT-P-0026693-T01-IM6-44, EVT-P-0043007-T01-IM6-71, EVT-P-0043704-T01-IM6-96, ... (6 total)
698	898	200	6	73	5	71	0.14084507042253522	0.21349536539409958	0.6706115482999183	EVT-P-0008101-T01-IM5-90, EVT-P-0017214-T01-IM6-23, EVT-P-0026693-T01-IM6-44, EVT-P-0043007-T01-IM6-71, EVT-P-0043704-T01-IM6-96, ... (6 total)
721	921	200	13	66	12	64	0.375	0.4213381555153707	0.3753692102786793	EVT-P-0005711-T02-IM6-2, EVT-P-0008764-T01-IM5-3, EVT-P-0017214-T01-IM6-23, EVT-P-0017924-T01-IM6-5, EVT-P-0024225-T01-IM6-37, ... (13 total)
735	935	200	13	66	12	64	0.375	0.4213381555153707	0.3753692102786793	EVT-P-0005711-T02-IM6-2, EVT-P-0008764-T01-IM5-3, EVT-P-0017214-T01-IM6-23, EVT-P-0017924-T01-IM6-5, EVT-P-0024225-T01-IM6-37, ... (13 total)
749	849	100	4	75	3	73	0.0821917808219178	0.14610705749946262	0.8353288055643661	EVT-P-0017214-T01-IM6-23, EVT-P-0026693-T01-IM6-44, EVT-P-0043007-T01-IM6-71, EVT-P-0043704-T01-IM6-96
799	849	50	4	75	3	73	0.0821917808219178	0.14610705749946262	0.8353288055643661	EVT-P-0017214-T01-IM6-23, EVT-P-0026693-T01-IM6-44, EVT-P-0043007-T01-IM6-71, EVT-P-0043704-T01-IM6-96
821	921	100	12	67	11	65	0.3384615384615385	0.3942133815551537	0.40426863770009347	EVT-P-0005711-T02-IM6-2, EVT-P-0008764-T01-IM5-3, EVT-P-0017214-T01-IM6-23, EVT-P-0017924-T01-IM6-5, EVT-P-0024225-T01-IM6-37, ... (12 total)
824	849	25	4	75	3	73	0.0821917808219178	0.14610705749946262	0.8353288055643661	EVT-P-0017214-T01-IM6-23, EVT-P-0026693-T01-IM6-44, EVT-P-0043007-T01-IM6-71, EVT-P-0043704-T01-IM6-96

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
848	873	25	4	75	3	73	0.0821917808219178	0.14610705749946262	0.8353288055643661	EVT-P-0017214-T01-IM6-23, EVT-P-0026693-T01-IM6-44, EVT-P-0043007-T01-IM6-71, EVT-P-0043704-T01-IM6-96
848	898	50	4	75	3	73	0.0821917808219178	0.14610705749946262	0.8353288055643661	EVT-P-0017214-T01-IM6-23, EVT-P-0026693-T01-IM6-44, EVT-P-0043007-T01-IM6-71, EVT-P-0043704-T01-IM6-96
848	948	100	12	67	11	65	0.3384615384615385	0.3942133815551537	0.40426863770009347	EVT-P-0005711-T02-IM6-2, EVT-P-0008764-T01-IM5-3, EVT-P-0017214-T01-IM6-23, EVT-P-0017924-T01-IM6-5, EVT-P-0024225-T01-IM6-37, ... (12 total)
848	1048	200	12	67	11	65	0.3384615384615385	0.3942133815551537	0.40426863770009347	EVT-P-0005711-T02-IM6-2, EVT-P-0008764-T01-IM5-3, EVT-P-0017214-T01-IM6-23, EVT-P-0017924-T01-IM6-5, EVT-P-0024225-T01-IM6-37, ... (12 total)
849	949	100	9	70	8	68	0.23529411764705882	0.3077808267681686	0.5117584380722447	EVT-P-0005711-T02-IM6-2, EVT-P-0008764-T01-IM5-3, EVT-P-0017924-T01-IM6-5, EVT-P-0024225-T01-IM6-37, EVT-P-0027991-T01-IM6-94, ... (9 total)
849	1049	200	9	70	8	68	0.23529411764705882	0.3077808267681686	0.5117584380722447	EVT-P-0005711-T02-IM6-2, EVT-P-0008764-T01-IM5-3, EVT-P-0017924-T01-IM6-5, EVT-P-0024225-T01-IM6-37, EVT-P-0027991-T01-IM6-94, ... (9 total)
871	921	50	8	71	8	68	None	1.0	-0.0	EVT-P-0005711-T02-IM6-2, EVT-P-0008764-T01-IM5-3, EVT-P-0017924-T01-IM6-5, EVT-P-0024225-T01-IM6-37, EVT-P-0027991-T01-IM6-94, ... (8 total)
896	921	25	8	71	8	68	None	1.0	-0.0	EVT-P-0005711-T02-IM6-2, EVT-P-0008764-T01-IM5-3, EVT-P-0017924-T01-IM6-5, EVT-P-0024225-T01-IM6-37, EVT-P-0027991-T01-IM6-94, ... (8 total)
921	946	25	8	71	8	68	None	1.0	-0.0	EVT-P-0005711-T02-IM6-2, EVT-P-0008764-T01-IM5-3, EVT-P-0017924-T01-IM6-5, EVT-P-0024225-T01-IM6-37, EVT-P-0027991-T01-IM6-94, ... (8 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
921	971	50	8	71	8	68	None	1.0	-0.0	EVT-P-0005711-T02-IM6-2, EVT-P-0008764-T01-IM5-3, EVT-P-0017924-T01-IM6-5, EVT-P-0024225-T01-IM6-37, EVT-P-0027991-T01-IM6-94, ... (8 total)
921	1021	100	8	71	8	68	None	1.0	-0.0	EVT-P-0005711-T02-IM6-2, EVT-P-0008764-T01-IM5-3, EVT-P-0017924-T01-IM6-5, EVT-P-0024225-T01-IM6-37, EVT-P-0027991-T01-IM6-94, ... (8 total)
921	1121	200	8	71	8	68	None	1.0	-0.0	EVT-P-0005711-T02-IM6-2, EVT-P-0008764-T01-IM5-3, EVT-P-0017924-T01-IM6-5, EVT-P-0024225-T01-IM6-37, EVT-P-0027991-T01-IM6-94, ... (8 total)

top_windows

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
550	575	25	56	23	56	20	None	0.022395326192794548	1.6498426075796853	EVT-P-0000208-T01-IM3-51, EVT-P-0000252-T01-IM3-54, EVT-P-0001010-T02-IM5-78, EVT-P-0002694-T02-IM5-12, EVT-P-0002873-T01-IM3-13, ... (56 total)
576	676	100	9	70	7	69	0.050724637681159424	0.03292909622023547	1.4824201888736057	EVT-P-0003125-T01-IM5-26, EVT-P-0013825-T01-IM5-17, EVT-P-0018069-T01-IM6-27, EVT-P-0020273-T02-IM6-29, EVT-P-0032407-T01-IM6-48, ... (9 total)
576	776	200	11	68	9	67	0.06716417910447761	0.04938099874808736	1.3064401305564408	EVT-P-0003125-T01-IM5-26, EVT-P-0008101-T01-IM5-90, EVT-P-0013825-T01-IM5-17, EVT-P-0018069-T01-IM6-27, EVT-P-0020273-T02-IM6-29, ... (11 total)
538	638	100	64	15	63	13	9.692307692307692	0.09073205275736923	1.04223926336373	EVT-P-0000208-T01-IM3-51, EVT-P-0000252-T01-IM3-54, EVT-P-0001010-T02-IM5-78, EVT-P-0002694-T02-IM5-12, EVT-P-0002873-T01-IM3-13, ... (64 total)
824	849	25	4	75	3	73	0.0821917808219178	0.14610705749946262	0.8353288055643661	EVT-P-0017214-T01-IM6-23, EVT-P-0026693-T01-IM6-44, EVT-P-0043007-T01-IM6-71, EVT-P-0043704-T01-IM6-96

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
569	594	25	58	21	57	19	6.0	0.1708418164114367	0.7674058197477294	EVT-P-0000208-T01-IM3-51, EVT-P-0000252-T01-IM3-54, EVT-P-0001010-T02-IM5-78, EVT-P-0002694-T02-IM5-12, EVT-P-0002873-T01-IM3-13, ... (58 total)
551	576	25	57	22	56	20	5.6	0.18597857838364168	0.7305370763495507	EVT-P-0000208-T01-IM3-51, EVT-P-0000252-T01-IM3-54, EVT-P-0001010-T02-IM5-78, EVT-P-0002694-T02-IM5-12, EVT-P-0002873-T01-IM3-13, ... (57 total)
649	849	200	6	73	5	71	0.14084507042253522	0.21349536539409958	0.6706115482999183	EVT-P-0008101-T01-IM5-90, EVT-P-0017214-T01-IM6-23, EVT-P-0026693-T01-IM6-44, EVT-P-0043007-T01-IM6-71, EVT-P-0043704-T01-IM6-96, ... (6 total)
614	639	25	7	72	6	70	0.17142857142857143	0.2458174736655749	0.6093872490052457	EVT-P-0018069-T01-IM6-27, EVT-P-0020273-T02-IM6-29, EVT-P-0032407-T01-IM6-48, EVT-P-0039474-T01-IM6-65, EVT-P-0048968-T01-IM6-77, ... (7 total)
589	639	50	8	71	7	69	0.2028985507246377	0.277241745596176	0.5571413753695974	EVT-P-0013825-T01-IM5-17, EVT-P-0018069-T01-IM6-27, EVT-P-0020273-T02-IM6-29, EVT-P-0032407-T01-IM6-48, EVT-P-0039474-T01-IM6-65, ... (8 total)
598	698	100	8	71	7	69	0.2028985507246377	0.277241745596176	0.5571413753695974	EVT-P-0008101-T01-IM5-90, EVT-P-0018069-T01-IM6-27, EVT-P-0020273-T02-IM6-29, EVT-P-0032407-T01-IM6-48, EVT-P-0039474-T01-IM6-65, ... (8 total)
635	735	100	9	70	8	68	0.23529411764705882	0.3077808267681686	0.5117584380722447	EVT-P-0008101-T01-IM5-90, EVT-P-0018069-T01-IM6-27, EVT-P-0020273-T02-IM6-29, EVT-P-0032407-T01-IM6-48, EVT-P-0039474-T01-IM6-65, ... (9 total)
849	949	100	9	70	8	68	0.23529411764705882	0.3077808267681686	0.5117584380722447	EVT-P-0005711-T02-IM6-2, EVT-P-0008764-T01-IM5-3, EVT-P-0017924-T01-IM6-5, EVT-P-0024225-T01-IM6-37, EVT-P-0027991-T01-IM6-94, ... (9 total)
594	794	200	10	69	9	67	0.26865671641791045	0.3374473627638185	0.47179396163015375	EVT-P-0008101-T01-IM5-90, EVT-P-0013825-T01-IM5-17, EVT-P-0018069-T01-IM6-27, EVT-P-0020273-T02-IM6-29, EVT-P-0032407-T01-IM6-48, ... (10 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
821	921	100	12	67	11	65	0.3384615384615385	0.3942133815551537	0.40426863770009347	EVT-P-0005711-T02-IM6-2, EVT-P-0008764-T01-IM5-3, EVT-P-0017214-T01-IM6-23, EVT-P-0017924-T01-IM6-5, EVT-P-0024225-T01-IM6-37, ... (12 total)
535	735	200	67	12	65	11	2.9545454545454546	0.3942133815551537	0.40426863770009347	EVT-P-0000208-T01-IM3-51, EVT-P-0000252-T01-IM3-54, EVT-P-0001010-T02-IM5-78, EVT-P-0002694-T02-IM5-12, EVT-P-0002873-T01-IM3-13, ... (67 total)
498	698	200	66	13	64	12	2.6666666666666665	0.4213381555153707	0.3753692102786793	EVT-P-0000208-T01-IM3-51, EVT-P-0000252-T01-IM3-54, EVT-P-0001010-T02-IM5-78, EVT-P-0002694-T02-IM5-12, EVT-P-0002873-T01-IM3-13, ... (66 total)
721	921	200	13	66	12	64	0.375	0.4213381555153707	0.3753692102786793	EVT-P-0005711-T02-IM6-2, EVT-P-0008764-T01-IM5-3, EVT-P-0017214-T01-IM6-23, EVT-P-0017924-T01-IM6-5, EVT-P-0024225-T01-IM6-37, ... (13 total)
539	639	100	65	14	63	13	2.423076923076923	0.44764096662830843	0.3490701751229587	EVT-P-0000208-T01-IM3-51, EVT-P-0000252-T01-IM3-54, EVT-P-0001010-T02-IM5-78, EVT-P-0002694-T02-IM5-12, EVT-P-0002873-T01-IM3-13, ... (65 total)
613	638	25	6	73	6	70	None	1.0	-0.0	EVT-P-0020273-T02-IM6-29, EVT-P-0032407-T01-IM6-48, EVT-P-0039474-T01-IM6-65, EVT-P-0048968-T01-IM6-77, EVT-P-0052861-T02-IM6-80, ... (6 total)
896	921	25	8	71	8	68	None	1.0	-0.0	EVT-P-0005711-T02-IM6-2, EVT-P-0008764-T01-IM5-3, EVT-P-0017924-T01-IM6-5, EVT-P-0024225-T01-IM6-37, EVT-P-0027991-T01-IM6-94, ... (8 total)
588	638	50	7	72	7	69	None	1.0	-0.0	EVT-P-0013825-T01-IM5-17, EVT-P-0020273-T02-IM6-29, EVT-P-0032407-T01-IM6-48, EVT-P-0039474-T01-IM6-65, EVT-P-0048968-T01-IM6-77, ... (7 total)
648	848	200	5	74	5	71	None	1.0	-0.0	EVT-P-0008101-T01-IM5-90, EVT-P-0017214-T01-IM6-23, EVT-P-0026693-T01-IM6-44, EVT-P-0043007-T01-IM6-71, EVT-P-0064951-T01-IM7-92

Per-event results (results.tsv)

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0062804-T03-IM7-1	P-0062804-T03-IM7	P-0062804	Soft Tissue Sarcoma	IMT	KIF5B-ALK	ALK	in-frame	five_prime	32304833	24	921	True	920	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - ALK (NM_004304) fusion: t(2;10)(p23.2;p11.22)(chr2:g.29447791::chr10:g.32304833) | Protein Fusion: in frame {KIF5B:ALK}	NA
EVT-P-0005711-T02-IM6-2	P-0005711-T02-IM6	P-0005711	Non-Small Cell Lung Cancer	NSCLCPD	KIF5B-ALK	ALK	in-frame	five_prime	32304765	24	921	True	920	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - ALK (NM_004304) fusion (KIF5B exons 1-24 fused with ALK exons 19-29) : t(2;10)(p23.2;p11.22)(chr2:g.29449548::chr10:g.32304765) | Protein fusion: in frame (KIF5B-ALK)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0008764-T01-IM5-3	P-0008764-T01-IM5	P-0008764	Non-Small Cell Lung Cancer	LUAD	KIF5B-ALK	ALK	unknown	five_prime	32305750	24	921	True	920	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - ALK (NM_004304) fusion (KIF5B exons 1-24 fused with ALK exons 19-29) : t(2;10)(p23.2;p11.22)(chr2:g.29448327::chr10:g.32305750) | Protein fusion: mid-exon (KIF5B-ALK)	NA
EVT-P-0008764-T02-IM6-4	P-0008764-T02-IM6	P-0008764	Non-Small Cell Lung Cancer	LUAD	KIF5B-ALK	ALK	unknown	five_prime	32305750	24	921	True	920	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - ALK (NM_004304) Fusion (KIF5B exons 1-24 fused with ALK exons 19-29) : t(2;10)(p23.2;p11.22)(chr2:g.29448327::chr10:g.:32305750) | Protein fusion: mid-exon (KIF5B-ALK)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0017924-T01-IM6-5	P-0017924-T01-IM6	P-0017924	Non-Small Cell Lung Cancer	LUAD	KIF5B-ALK	ALK	in-frame	five_prime	32305310	24	921	True	920	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - ALK (NM_004304) fusion: t(2;10)(p23.2;p11.22)(chr2:g.29447945::chr10:g.32305310) | Protein Fusion: in frame {KIF5B:ALK}	NA
EVT-P-0052330-T01-IM6-6	P-0052330-T01-IM6	P-0052330	Ampullary Cancer	AMPCA	KIF5B-ARID5B	ARID5B	unknown	five_prime	32312962	15	575	True	575	True	exact match	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - ARID5B (NM_032199) rearrangement: c.1726-998:KIF5B_c.2475:ARID5Binv | Protein Fusion: mid-exon {KIF5B:ARID5B}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0021038-T01-IM6-7	P-0021038-T01-IM6	P-0021038	Non-Small Cell Lung Cancer	LUAD	KIF5B-EGFR	EGFR	in-frame	five_prime	32312089	15	575	True	575	True	exact match	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - EGFR (NM_005228) rearrangement: t(7;10)(p11.2;q11.22)(chr7:g.55241486::chr10:g.32312089) | Protein Fusion: in frame {KIF5B:EGFR}	NA
EVT-P-0033090-T01-IM6-8	P-0033090-T01-IM6	P-0033090	Non-Small Cell Lung Cancer	LUAD	KIF5B-MET	MET	in-frame	five_prime	32305234	24	921	True	920	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - MET (NM_000245) fusion (KIF5B exons 1-24 fused to MET exons 15-21):t(7;10)(q31.2;p11.22)(chr7:g.116412397::chr10:g.32305234) | Protein Fusion: in frame {KIF5B:MET}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0043854-T01-IM6-9	P-0043854-T01-IM6	P-0043854	Non-Small Cell Lung Cancer	LUAD	KIF5B-MET	MET	in-frame	five_prime	32304904	24	921	True	920	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - MET (NM_000245) fusion: t(7;10)(q31.2;p11.22)(chr7:g.116412098::chr10:g.32304904) | Protein Fusion: in frame {KIF5B:MET}	NA
EVT-P-0000208-T02-IM5-10	P-0000208-T02-IM5	P-0000208	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	RET	in-frame	five_prime	32313935	15	575	True	575	True	exact match	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused with RET exons 12- 20): c.1726-1971:KIF5B_c.2137-385:RETinv | Protein fusion: in frame (KIF5B-RET)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0008289-T01-IM5-11	P-0008289-T01-IM5	P-0008289	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	RET	in-frame	five_prime	32312788	15	575	True	575	True	exact match	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20) : c.1726-824:KIF5B_c.2137-284:RETinv | Protein fusion: in frame (KIF5B-RET)	NA
EVT-P-0002694-T02-IM5-12	P-0002694-T02-IM5	P-0002694	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	RET	in-frame	five_prime	32314319	15	575	True	575	True	exact match	retained	1.0	False	Kinesin motor domain			NA | Protein fusion: in frame (KIF5B-RET)	NA
EVT-P-0002873-T01-IM3-13	P-0002873-T01-IM3	P-0002873	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	RET	in-frame	five_prime	32312502	15	575	True	575	True	exact match	retained	1.0	False	Kinesin motor domain			NA | Protein fusion: in frame (KIF5B-RET)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0009514-T01-IM5-14	P-0009514-T01-IM5	P-0009514	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	RET	in-frame	five_prime	32313425	15	575	True	575	True	exact match	retained	1.0	False	Kinesin motor domain			Note: The KIF5B (NM_004521) - RET (NM_020975) reciprocal rearrangement event is an inversion which results in the fusion of KIF5B exons 1 to 15 and RET exons 12 to 20. The clinical trial research is currently ongoing with cabozantinib. PMID: 23533264 | Protein fusion: in frame (KIF5B-RET)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0011035-T01-IM5-15	P-0011035-T01-IM5	P-0011035	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	RET	in-frame	five_prime	32314260	15	575	True	575	True	exact match	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused with RET exons 12-20) : c.1726-2296:KIF5B_c.2137-710:RETinv | Protein fusion: in frame (KIF5B-RET)	NA
EVT-P-0012648-T01-IM5-16	P-0012648-T01-IM5	P-0012648	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	RET	in-frame	five_prime	32315493	15	575	True	575	True	exact match	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1863:KIF5B_c.2137-335:RETinv | Protein fusion: in frame (KIF5B-RET)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0013825-T01-IM5-17	P-0013825-T01-IM5	P-0013825	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	RET	unknown	five_prime	32311908	16	594	False	594	True	exact match	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-16 fused to RET exons 12-20): c.1782:KIF5B_c.2137-512:RETinv | Protein fusion: mid-exon (KIF5B-RET)	NA
EVT-P-0014808-T01-IM6-18	P-0014808-T01-IM6	P-0014808	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	RET	in-frame	five_prime	32315591	15	575	True	575	True	exact match	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+1765:KIF5B_c.2136+871:RETinv | Protein fusion: inframe (KIF5B-RET)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0014808-T02-IM6-19	P-0014808-T02-IM6	P-0014808	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	RET	in-frame	five_prime	32315592	15	575	True	575	True	exact match	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+1765:KIF5B_c.2136+871:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0014808-T03-IM6-20	P-0014808-T03-IM6	P-0014808	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	RET	in-frame	five_prime	32315592	15	575	True	575	True	exact match	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1764:KIF5B_c.2136+872:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0016221-T01-IM6-21	P-0016221-T01-IM6	P-0016221	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	RET	in-frame	five_prime	32317081	15	575	True	575	True	exact match	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused in-frame to RET exons 12-20) : c.1725+275:KIF5B_c.2137-764:RETinv | Protein Fusion: in frame {KIF5B: RET}	NA
EVT-P-0016221-T02-IM6-22	P-0016221-T02-IM6	P-0016221	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	RET	in-frame	five_prime	32317081	15	575	True	575	True	exact match	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 with RET exons 12-20 including the kinase domain): c.1725+275:KIF5B_c.2137-764:RETinv | Protein Fusion: in frame {KIF5B: RET}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0017214-T01-IM6-23	P-0017214-T01-IM6	P-0017214	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	RET	in-frame	five_prime	32306314	23	848	True	848	True	exact match	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-23 fused in-frame with RET exons 12-20): c.2545-27:KIF5B_c.2136+463:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0017214-T02-IM6-24	P-0017214-T02-IM6	P-0017214	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	RET	in-frame	five_prime	32306314	23	848	True	848	True	exact match	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-23 fused to RET exons 12-20): c.2545-27:KIF5B_c.2136+463:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0017523-T02-IM6-25	P-0017523-T02-IM6	P-0017523	Non-Small Cell Lung Cancer	LUPC	KIF5B-RET	RET	in-frame	five_prime	32314326	15	575	True	575	True	exact match	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1726-2362:KIF5B_c.2137-840:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0003125-T01-IM5-26	P-0003125-T01-IM5	P-0003125	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	RET	in-frame	three_prime	32315549	16	576	True	576	True	exact match	lost	0.0	False		Kinesin motor domain		NA | Protein fusion: in frame (RET-KIF5B)	NA
EVT-P-0018069-T01-IM6-27	P-0018069-T01-IM6	P-0018069	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	RET	out-of-frame	three_prime	32311257	17	639	True	639	True	exact match	lost	0.0	False		Kinesin motor domain		RET (NM_020975) - KIF5B (NM_004521) rearrangement: c.1648+160:RET_c.1915-72:KIF5Binv | Protein Fusion: out of frame {RET:KIF5B}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0019535-T02-IM6-28	P-0019535-T02-IM6	P-0019535	Non-Small Cell Lung Cancer	LUNE	KIF5B-RET	RET	in-frame	five_prime	32314077	15	575	True	575	True	exact match	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-16 fused to RET exons 12-20): c.1726-2113:KIF5B_c.2137-119:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0020273-T02-IM6-29	P-0020273-T02-IM6	P-0020273	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	RET	in-frame	five_prime	32311645	16	638	True	638	True	exact match	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-16 fused to RET exons 12-20): c.1914+131:KIF5B_c.2137-213:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0003125-T01-IM5-30	P-0003125-T01-IM5	P-0003125	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	RET	in-frame	five_prime	32315561	15	575	True	575	True	exact match	retained	1.0	False	Kinesin motor domain			NA | Protein fusion: in frame (KIF5B-RET)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0021150-T01-IM6-31	P-0021150-T01-IM6	P-0021150	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	RET	in-frame	five_prime	32317044	15	575	True	575	True	exact match	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+312:KIF5B_c.2137-441:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0021578-T01-IM6-32	P-0021578-T01-IM6	P-0021578	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	RET	in-frame	five_prime	32316137	15	575	True	575	True	exact match	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+1231:KIF5B_c.2136+624:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0021879-T01-IM6-33	P-0021879-T01-IM6	P-0021879	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	RET	in-frame	five_prime	32312846	15	575	True	575	True	exact match	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B1 exons 1-15 fused with RET exons 12-20 including the kinase domain.): c.1726-882:KIF5B_c.2136+515_RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0021937-T01-IM6-34	P-0021937-T01-IM6	P-0021937	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	RET	in-frame	five_prime	32316470	15	575	True	575	True	exact match	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused with RET exons 12-20) : c.1725+886:KIF5B_c.2137-477:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0021937-T03-IM6-35	P-0021937-T03-IM6	P-0021937	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	RET	in-frame	five_prime	32316470	15	575	True	575	True	exact match	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+886:KIF5B_c.2137-477:RET inv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0022866-T01-IM6-36	P-0022866-T01-IM6	P-0022866	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	RET	in-frame	five_prime	32312731	15	575	True	575	True	exact match	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - RET (NM_020975) Fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1726-767:KIF5B_c.2137-161:RET inv | Protein Fusion: in frame {KIF5B:RET}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0024225-T01-IM6-37	P-0024225-T01-IM6	P-0024225	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	RET	unknown	five_prime	32305017	24	921	True	920	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-24 fused to RET exons 10-20): c.2762-430:KIF5B_c.1866:RETinv | Protein Fusion: mid-exon {KIF5B:RET}	NA
EVT-P-0024843-T01-IM6-38	P-0024843-T01-IM6	P-0024843	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	RET	in-frame	five_prime	32313303	15	575	True	575	True	exact match	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 with RET exons 12-20): c.1726-1339:KIF5B_c.2137-299:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0025100-T01-IM6-39	P-0025100-T01-IM6	P-0025100	Non-Small Cell Lung Cancer	NSCLCPD	KIF5B-RET	RET	in-frame	five_prime	32314237	15	575	True	575	True	exact match	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1726-2273:KIF5B_c.2137-521:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0025100-T02-IM6-40	P-0025100-T02-IM6	P-0025100	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	RET	in-frame	five_prime	32314237	15	575	True	575	True	exact match	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) and RET (NM_020975) Fusion (KIF5B exon 16 fused to RET exon 12): c.1726-2273:KIF5B_c.2137-788:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0025838-T01-IM6-41	P-0025838-T01-IM6	P-0025838	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	RET	in-frame	five_prime	32312565	15	575	True	575	True	exact match	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused with RET exons 12-20.): c.1726-601:KIF5B_c.2136+788:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0025928-T01-IM6-42	P-0025928-T01-IM6	P-0025928	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	RET	in-frame	five_prime	32316873	15	575	True	575	True	exact match	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+483:KIF5B_c.2137-122C>T:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0025928-T02-IM6-43	P-0025928-T02-IM6	P-0025928	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	RET	in-frame	five_prime	32316873	15	575	True	575	True	exact match	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+483:KIF5B_c.2137-122:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0026693-T01-IM6-44	P-0026693-T01-IM6	P-0026693	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	RET	in-frame	five_prime	32306824	23	848	True	848	True	exact match	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - RET (NM_020975) fusion:c.2544+156:KIF5_c.2136+327:RETinv. | Protein Fusion: in frame {KIF5B:RET}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0026777-T01-IM6-45	P-0026777-T01-IM6	P-0026777	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	RET	in-frame	five_prime	32313518	15	575	True	575	True	exact match	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-16 fused to RET exons 12-20): c.1726-1554:KIF5B_c.2136+533:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0026777-T02-IM6-46	P-0026777-T02-IM6	P-0026777	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	RET	in-frame	five_prime	32313518	15	575	True	575	True	exact match	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-16 fused to RET exons 12-20): c.1726-1554:KIF5B_c.2136+533:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0027723-T01-IM6-47	P-0027723-T01-IM6	P-0027723	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	RET	unknown	five_prime	32316390	15	575	True	575	True	exact match	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 11-20):c.1725+966:KIF5B_c.2123:RETinv | Protein Fusion: mid-exon {KIF5B:RET}	NA
EVT-P-0032407-T01-IM6-48	P-0032407-T01-IM6	P-0032407	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	RET	in-frame	five_prime	32311546	16	638	True	638	True	exact match	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1914+230:KIF5B_c.2137-302:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0030453-T01-IM6-49	P-0030453-T01-IM6	P-0030453	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	RET	in-frame	five_prime	32314360	15	575	True	575	True	exact match	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-2396:KIF5B_c.2136+727:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0032227-T01-IM6-50	P-0032227-T01-IM6	P-0032227	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	RET	in-frame	five_prime	32315356	15	575	True	575	True	exact match	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+2000:KIF5B_c.2137-907:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0000208-T01-IM3-51	P-0000208-T01-IM3	P-0000208	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	RET	in-frame	five_prime	32313935	15	575	True	575	True	exact match	retained	1.0	False	Kinesin motor domain			RET (NM_020975) Inversion (c.2137-385_KIF5B:c.1726-1971inv) | Protein fusion: in frame (KIF5B-RET)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0032686-T01-IM6-52	P-0032686-T01-IM6	P-0032686	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	RET	in-frame	five_prime	32315057	15	575	True	575	True	exact match	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+2299:KIF5B_c.2137-50:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0032753-T01-IM6-53	P-0032753-T01-IM6	P-0032753	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	RET	in-frame	five_prime	32316033	15	575	True	575	True	exact match	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1323:KIF5B_c.2136+741:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0000252-T01-IM3-54	P-0000252-T01-IM3	P-0000252	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	RET	in-frame	five_prime	32313497	15	575	True	575	True	exact match	retained	1.0	False	Kinesin motor domain			RET (NM_020975) Inversion (c.2136+413_KIF5B:c.1726-153inv) | Protein fusion: in frame (KIF5B-RET)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0033700-T02-IM6-55	P-0033700-T02-IM6	P-0033700	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	RET	in-frame	five_prime	32315902	15	575	True	575	True	exact match	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1454:KIF5B_c.2137-362:RET | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0034357-T01-IM6-56	P-0034357-T01-IM6	P-0034357	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	RET	in-frame	five_prime	32315493	15	575	True	575	True	exact match	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1863:KIF5B_c.2137-845:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0036092-T01-IM6-57	P-0036092-T01-IM6	P-0036092	Non-Small Cell Lung Cancer	LNET	KIF5B-RET	RET	in-frame	five_prime	32316218	15	575	True	575	True	exact match	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+1138:KIF5B_c.2136+229:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0036637-T01-IM6-58	P-0036637-T01-IM6	P-0036637	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	RET	in-frame	five_prime	32313695	15	575	True	575	True	exact match	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1726-1731:KIF5B_c.2137-67:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0036737-T02-IM6-59	P-0036737-T02-IM6	P-0036737	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	RET	in-frame	five_prime	32315869	15	575	True	575	True	exact match	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+1487:KIF5B_c.2137-7:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0037367-T01-IM6-60	P-0037367-T01-IM6	P-0037367	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	RET	in-frame	five_prime	32316318	15	575	True	575	True	exact match	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1038:KIF5B_c.2137-158:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0037494-T01-IM6-61	P-0037494-T01-IM6	P-0037494	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	RET	in-frame	five_prime	32316371	15	575	True	575	True	exact match	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - RET (NM_020975) Fusion (KIF5B exon15 fused with RET exon12): c.1725+985:KIF5B_c.2137-277:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0038961-T01-IM6-62	P-0038961-T01-IM6	P-0038961	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	RET	in-frame	five_prime	32312257	15	575	True	575	True	exact match	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-293:KIF5B_c.2136+591:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0039257-T01-IM6-63	P-0039257-T01-IM6	P-0039257	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	RET	unknown	five_prime	32316224	15	575	True	575	True	exact match	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1132:KIF5B_c.2012:RETinv | Protein Fusion: mid-exon {KIF5B:RET}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0039257-T03-IM7-64	P-0039257-T03-IM7	P-0039257	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	RET	unknown	five_prime	32316224	15	575	True	575	True	exact match	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1132:KIF5B_c.2012:RETinv | Protein Fusion: mid-exon {KIF5B:RET}	NA
EVT-P-0039474-T01-IM6-65	P-0039474-T01-IM6	P-0039474	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	RET	in-frame	five_prime	32311388	16	638	True	638	True	exact match	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1915-203:KIF5B_c.2136+175:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0039474-T02-IM6-66	P-0039474-T02-IM6	P-0039474	Non-Small Cell Lung Cancer	NSCLC	KIF5B-RET	RET	in-frame	five_prime	32311388	16	638	True	638	True	exact match	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1915-203:KIF5B_c.2136+175:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0039941-T01-IM6-67	P-0039941-T01-IM6	P-0039941	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	RET	in-frame	five_prime	32315316	15	575	True	575	True	exact match	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+2040:KIF5B_c.2137-693:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0042104-T01-IM6-68	P-0042104-T01-IM6	P-0042104	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	RET	in-frame	five_prime	32313020	15	575	True	575	True	exact match	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-1056:KIF5B_c.2136+321:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0042580-T01-IM6-69	P-0042580-T01-IM6	P-0042580	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	RET	in-frame	five_prime	32315737	15	575	True	575	True	exact match	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1619:KIF5B_c.2136+760:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0042608-T01-IM6-70	P-0042608-T01-IM6	P-0042608	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	RET	in-frame	five_prime	32313634	15	575	True	575	True	exact match	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-1670:KIF5B_c.2137-230:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0043007-T01-IM6-71	P-0043007-T01-IM6	P-0043007	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	RET	in-frame	five_prime	32306812	23	848	True	848	True	exact match	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.2544+168:KIF5B_c.2136+629:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0043007-T02-IM6-72	P-0043007-T02-IM6	P-0043007	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	RET	in-frame	five_prime	32306812	23	848	True	848	True	exact match	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.2544+168:KIF5B_c.2136+629:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0005573-T01-IM5-73	P-0005573-T01-IM5	P-0005573	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	RET	in-frame	five_prime	32312824	15	575	True	575	True	exact match	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - RET (NM_020975) Fusion (KIF5B exon15 fused with RET exon12): c.1726-860:KIF5B_c.2136+234:RETinv | Protein fusion: in frame (KIF5B-RET)	NA
EVT-P-0046726-T01-IM6-74	P-0046726-T01-IM6	P-0046726	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	RET	in-frame	five_prime	32316588	15	575	True	575	True	exact match	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+768:KIF5B_c.2136+803:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0046827-T02-IM6-75	P-0046827-T02-IM6	P-0046827	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	RET	in-frame	five_prime	32313269	15	575	True	575	True	exact match	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-1305:KIF5B_c.2136+705:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0048829-T01-IM6-76	P-0048829-T01-IM6	P-0048829	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	RET	in-frame	five_prime	32316408	15	575	True	575	True	exact match	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+948:KIF5B_c.2136+432:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0048968-T01-IM6-77	P-0048968-T01-IM6	P-0048968	Small Cell Lung Cancer	CSCLC	KIF5B-RET	RET	in-frame	five_prime	32311565	16	638	True	638	True	exact match	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1914+211:KIF5B_c.2137-223:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0001010-T02-IM5-78	P-0001010-T02-IM5	P-0001010	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	RET	in-frame	five_prime	32312744	15	575	True	575	True	exact match	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-780:KIF5B_c.2136+679:RETinv | Protein fusion: in frame (KIF5B-RET)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0052826-T01-IM6-79	P-0052826-T01-IM6	P-0052826	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	RET	in-frame	five_prime	32314322	15	575	True	575	True	exact match	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-2358:KIF5B_c.2137-850:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0052861-T02-IM6-80	P-0052861-T02-IM6	P-0052861	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	RET	in-frame	five_prime	32311205	16	638	True	638	True	exact match	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1915-20:KIF5B_c.2136+76:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0053175-T01-IM6-81	P-0053175-T01-IM6	P-0053175	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	RET	in-frame	five_prime	32312246	15	575	True	575	True	exact match	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-282:KIF5B_c.2137-870:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0055183-T01-IM6-82	P-0055183-T01-IM6	P-0055183	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	RET	in-frame	five_prime	32315558	15	575	True	575	True	exact match	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1798:KIF5B_c.2136+124:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0055183-T02-IM6-83	P-0055183-T02-IM6	P-0055183	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	RET	in-frame	five_prime	32315558	15	575	True	575	True	exact match	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1798:KIF5B_c.2136+124:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0057072-T02-IM6-84	P-0057072-T02-IM6	P-0057072	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	RET	in-frame	five_prime	32314848	15	575	True	575	True	exact match	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+2508:KIF5B_c.2136+922:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0057072-T03-IM7-85	P-0057072-T03-IM7	P-0057072	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	RET	in-frame	five_prime	32314848	15	575	True	575	True	exact match	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+2508:KIF5B_c.2136+922:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0059913-T01-IM7-86	P-0059913-T01-IM7	P-0059913	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	RET	unknown	five_prime	32314915	15	575	True	575	True	exact match	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+2441:KIF5B_c.2060:RETinv | Protein Fusion: mid-exon {KIF5B:RET}	NA
EVT-P-0060748-T02-IM7-87	P-0060748-T02-IM7	P-0060748	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	RET	in-frame	five_prime	32314268	15	575	True	575	True	exact match	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-2304:KIF5B_c.2137-505:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0060748-T03-IM7-88	P-0060748-T03-IM7	P-0060748	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	RET	in-frame	five_prime	32314268	15	575	True	575	True	exact match	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-2304:KIF5B_c.2137-505:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0061331-T01-IM7-89	P-0061331-T01-IM7	P-0061331	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	RET	in-frame	five_prime	32311471	16	638	True	638	True	exact match	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1915-286:KIF5B_c.2136+700:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0008101-T01-IM5-90	P-0008101-T01-IM5	P-0008101	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	RET	in-frame	five_prime	32310097	18	698	True	698	True	exact match	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-18 fused with RET exons 12-20) : c.2095-38:KIF5B_c.2136+271:RETinv | Protein fusion: in frame (KIF5B-RET)	NA
EVT-P-0062948-T03-IM7-91	P-0062948-T03-IM7	P-0062948	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	RET	in-frame	five_prime	32315731	15	575	True	575	True	exact match	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1625:KIF5B_c.2136+27:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA

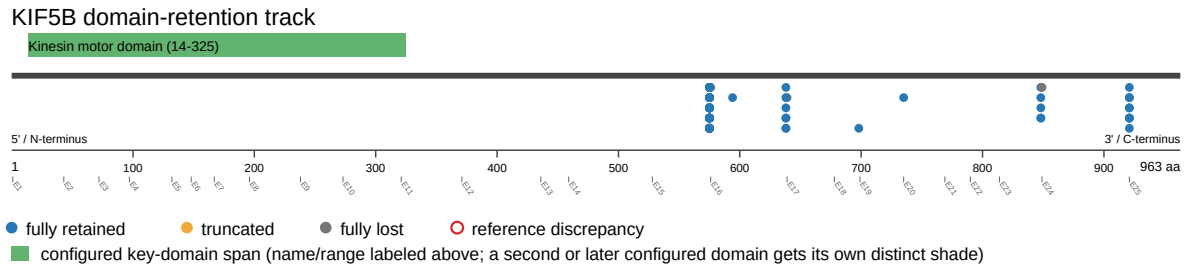
event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0064951-T01-IM7-92	P-0064951-T01-IM7	P-0064951	Cancer of Unknown Primary	CUP	KIF5B-RET	RET	out-of-frame	five_prime	32309948	19	735	True	734	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.2204+2:KIF5B_c.2137-902:RETinv | Protein Fusion: out of frame {KIF5B:RET}	NA
EVT-P-0065876-T01-IM7-93	P-0065876-T01-IM7	P-0065876	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	RET	in-frame	five_prime	32313866	15	575	True	575	True	exact match	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-1902:KIF5B_c.2137-733:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0027991-T01-IM6-94	P-0027991-T01-IM6	P-0027991	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	RET	in-frame	five_prime	32304835	24	921	True	920	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	Kinesin motor domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-24 fused to RET exons 9-20): c.2762-248:KIF5B_c.1649-100:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0004995-T01-IM5-95	P-0004995-T01-IM5	P-0004995	Non-Small Cell Lung Cancer	LUAD	KIF5B-RET	RET	in-frame	five_prime	32314330	15	575	True	575	True	exact match	retained	1.0	False	Kinesin motor domain			KIF5B(NM_004521)-RET(NM_020975) Fusion (KIF5B exon 15 fused to RET exon 12) : c.1726-2366:KIF5B_c.2136+476:RETinv | Protein fusion: in frame (KIF5B-RET)	NA

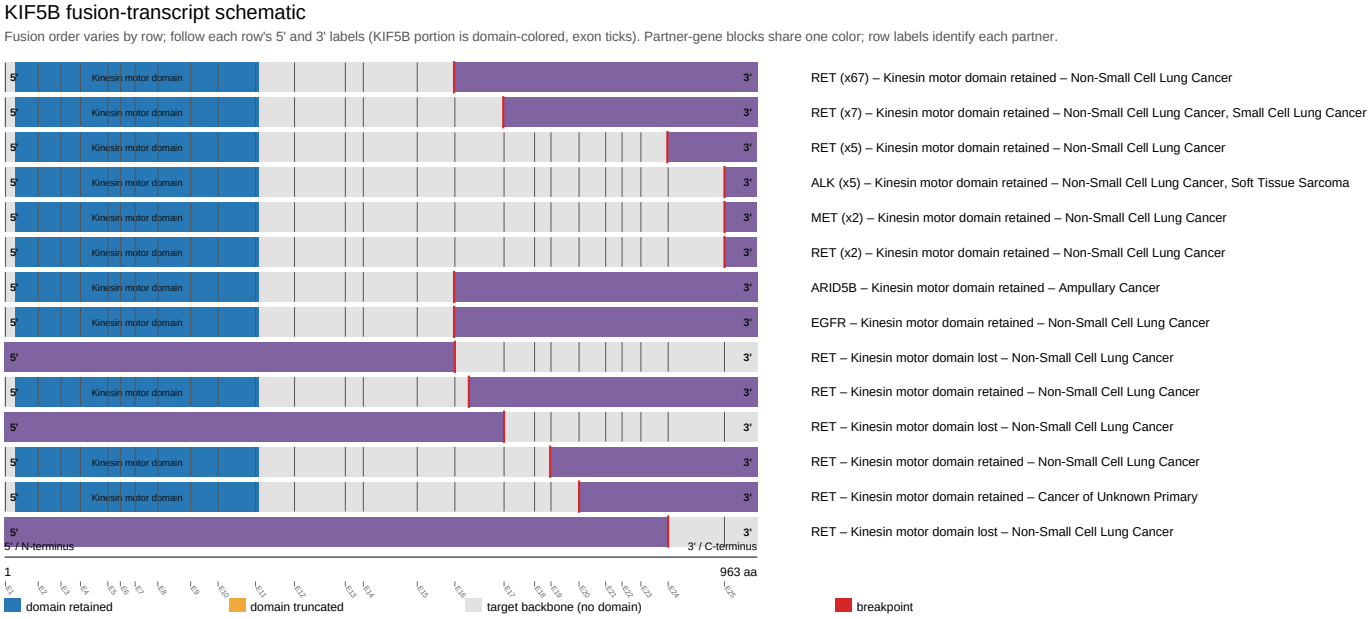
event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0043704-T01-IM6-96	P-0043704-T01-IM6	P-0043704	Non-Small Cell Lung Cancer	LUAD	RET-KIF5B	RET	in-frame	three_prime	32306364	24	849	True	849	True	exact match	lost	0.0	False		Kinesin motor domain		RET (NM_020975) - KIF5B (NM_004521) rearrangement: c.2136+147:RET_c.2545-77:KIF5Binv | Protein Fusion: in frame {RET:KIF5B}	NA

Figures

domain retention outliers



fusion schematic



Partner-block colors are decorative; row labels identify partners.